Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	*(a)*			continuously changes direction (1)	1			1		
	(b)	(i)		Substitution into $E = Pt$: 2 600 × 15 (1) = 39 000 [J] Total energy = 39 000 ecf × 6 (1) = 234 000 [J] (1) cao **Alternative** 15 × 6 (1) = 90 [s] $E = P\ t$ = 2 600 × 90 ecf (1) = 234 000 [J] (1) cao NB × 15 (1) × 6 (1) Answer 234 000 (1) 15 × 6 = 90 → 1 mark only Use of 230 V in calculation → 0 marks.	1	1 1		3	3	
		(ii)		Substitution: $\frac{2\,600}{230}$ (1) =11.3 [A] (1) [accept 11 A]	1	1		2	2	
		(iii)		same answer as (ii): expect live =11.3 **and** neutral = 11.3 (1) earth = 0 (1)	1	1		2		
	(c)			[Detects] {current / magnetic field} difference (1) ... . between the live and neutral [leads] (1)	2			2		
	(d)			½ power (1) same {number of kWh / number of units / energy} (1) Or by calculation 1300 W × 30 s (1) = 39 000 J + comparison with 2 600 × 15 [or 39 000](1) To award 2 marks 'claim is not true' [or equiv] must be seen.			2	2	1	
				Question 5 total	**6**	**4**	**2**	**12**	**6**	**0**